Date: ……

Practical No: 1: Execute Basic TCP/IP utilities and commands. (eg: ping, ipconfig, tracert, arp, tcpdump, whois, host, netstat, nslookup, ftp, telnet etc...)

A. Objective:

To utilize basic TCP/IP utilities and commands to troubleshoot network issues, gather configuration information, and analyze connectivity, aiding in effective network management and diagnostics.

B. Expected Program Outcomes (POs): PO1, PO2, PO5, PO8, PO11

C. Expected Skills to be developed based on competency:

The practical is expected to develop the following skills:

- **Network Troubleshooting**: Identify and fix connectivity issues using tools like ping and arp.
- **IP Configuratio**n: View and manage IP settings with ipconfig (Windows) or ifconfig (Linux).
- **Routing Analysis**: Trace packet routes using tracert (Windows) or traceroute (Linux).
- **ARP Management**: Understand and troubleshoot ARP for address resolution.
- **Traffic Analysis**: Capture and analyze network traffic with tcpdump.
- **Domain Information**: Retrieve domain and IP details using whois, host, and nslookup.
- **Connection Monitoring**: Monitor network connections and ports using netstat.
- **File Transfer**: Transfer files using FTP and understand basic telnet for remote connections.

D. Expected Course Outcomes (Cos): CO1

E. Practical Outcome (PRo)

Execute Basic TCP/IP utilities and commands.

F. Expected Affective domain Outcome (ADos)

1. Follow coding standards.
2. Demonstrate working as a leader/ a team member.
3. Follow ethical practices.

G. Prerequisite

Theory: TCP/IP

Model:

Explanation: The TCP/IP model is a conceptual framework that standardizes the functions of a telecommunication or computing system into different layers. It consists of the Application, Transport, Network, Data Link, and Physical layers.

Example: The process of sending an email involves multiple layers of the TCP/IP model, from the application layer (email client) to the physical layer (network hardware).

IP Addressing:

Explanation: IP addressing is a system used to uniquely identify devices on a network. IPv4 addresses are written as four sets of numbers, and IPv6 uses a longer hexadecimal format.

Example: IPv4 address - 192.168.1.1, IPv6 address - 3001:0db8:85a3:0000:0000:8a2e:0370:7334

DNS (Domain Name System):

Explanation: DNS translates human-readable domain names into IP addresses. It consists of a hierarchical system of DNS servers.

Example: Resolving the domain "www.example.com" to the IP address 203.0.113.5 using DNS.

Routing:

Explanation: Routing is the process of forwarding data packets between different networks. Routers use routing tables to determine the best path for packet delivery.

Example: A router deciding the best path for a packet to travel from one network to another based on the destination IP address.

Network Protocols:

Explanation: Network protocols define rules and conventions for communication between devices on a network.

Example: TCP ensures reliable, connection-oriented communication, while UDP provides faster but less reliable communication for activities like streaming.

Ethernet and MAC Addresses:

Explanation: Ethernet is a widely used LAN technology, and MAC addresses are unique identifiers assigned to network interfaces.

Example: A MAC address like "00:1A:2B:3C:4D:5E" uniquely identifies a network device.

Firewall Concepts:

Explanation: Firewalls are network security devices that control incoming and outgoing network traffic based on an organization's predefined security rules.

Example: Configuring a firewall to allow or block specific ports for incoming and outgoing traffic.

FTP and Telnet Protocols:

Explanation: FTP is a protocol for transferring files between computers, while Telnet provides a text-based interface for remote access.

Example: Using FTP to upload/download files to/from a server or using Telnet to remotely access and manage a router.

Network Security Basics:

Explanation: Network security involves protecting data integrity, confidentiality, and availability. It includes measures like encryption, access controls, and regular security audits.

Example: Implementing WPA2/WPA3 encryption for Wi-Fi networks to secure wireless communication.

Command-Line Interface (CLI):

Explanation: CLI is a text-based interface where users interact with a computer by typing commands. It is efficient for network configuration and troubleshooting.

Example: Using the Command Prompt (Windows) or Terminal (Linux) to execute commands like ping or ipconfig/ifconfig.

Network Topology:

Explanation: Network topology defines the physical or logical layout of interconnected devices in a network.

Example: A star topology where all devices are connected to a central hub or a bus topology where devices share a common communication line.

Network

Commands

ping

Explanation: Tests reachability to a host by sending ICMP echo request packets.

Example: ping www.google.com

ipconfig (Windows) / ifconfig (Linux)

Explanation: Displays IP configuration details for network interfaces.

Example: ipconfig (Windows) or ifconfig (Linux)

tracert (Windows) / traceroute (Linux)

Explanation: Traces the route that packets take to reach a destination.

Example: tracert www.example.com (Windows) or

traceroute www.example.com (Linux)

arp

Explanation: Displays and modifies the ARP (Address Resolution Protocol) cache.

Example: arp -a

tcpdump

Explanation: Captures and analyzes network traffic.

Example: tcpdump -i eth0

whois

Explanation: Retrieves information about a domain or IP address from the WHOIS database.

Example: whois www.example.com

host

Explanation: Performs DNS lookups to translate domain names to IP addresses and vice versa.

Example: host www.example.com

netstat

Explanation: Displays network connections, routing tables, interface statistics, etc.

Example: netstat -a

nslookup

Explanation: Queries DNS servers to obtain domain name or IP address information.

Example: nslookup www.example.com

ftp

Explanation: Connects to an FTP server and transfers files.

Example: ftp ftp.example.com

telnet

Explanation: Connects to a remote host using the Telnet protocol.

Example: telnet www.example.com 80

H. Resources/Equipment Required

Sr. No.	Instrument/Equipment /Components/Trainer kit	Specification	Quantity
1	Hardware: Computer System	Computer (i3-i5 preferable),RAM minimum 2 GB and onwards	As per batch size
2	Software: Operating System	Windows/ Linux	
3	Software	Command-Line Interface (CLI) ● For Windows: Command Prompt or PowerShell. ● For Linux: Terminal.	

I. Safety and necessary Precautions followed

- Exercise caution, ensure proper authorization, and adhere to ethical standards when performing network commands, respecting privacy and security protocols to avoid unintended disruptions or unauthorized access.

J. Procedure to be followed/Source code:

K. Practical related Quiz.

1. Which command is used to display the IP configuration for all network interfaces on a Windows system?
   a) ping
   b) tracert
   c) ipconfig
   d) nslookup
2. What is the purpose of the "ping" command?
   a) Trace the route to a destination
   b) Display IP configuration
   c) Test the reachability of a host on an Internet Protocol (IP) network
   d) Display network connections and listening ports
3. Which command can be used to view the current routing table on a computer?
   a) netstat
   b) arp
   c) host
   d) ftp
4. What does the "traceroute" (or "tracert" on Windows) command do?
   a) Display the IP configuration
   b) Display the ARP cache
   c) Trace the route that packets take to reach a destination
   d) Test the reachability of a host using ICMP
5. Which command can be used to find the IP address associated with a domain name?
   a) ftp
   b) whois
   c) nslookup
   d) telnet

L. References / Suggestions (Lab manual designer should give)

1. https://www.javatpoint.com/linux-networking-commands
2. https://www.geeksforgeeks.org/c ommand-line-tools-and-utilities-for-network-management-in-linux/
3. https://www.youtube.com/watch?v=rurs7cdT5cc